4 **(a)** State and explain the **two** features of a liquid-in-glass thermometer that are necessary for linearity.

statement 1 .....

explanation .....

statement 2 .....

explanation .....
[4]

(b) The value of the heat capacity of the hot junction of a thermocouple thermometer is important in ensuring that it can measure temperature changes very rapidly.

Explain why.

.....

.....

.....

..... [2]

(c) The hot junction of a thermocouple thermometer has a heat capacity of 0.11 J / °C.

Calculate the thermal energy required to increase the temperature of the hot junction from 20 °C to 345 °C.

energy = ... [3]

[Total: 9]

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